

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade) — ANSWER KEY (Version 1)

Syllabus: Units 1-6 & 9 (Comprehensive)

Total Marks: 65

Prepared for Marking Use

Version: 1

SECTION A — MCQ Answer Key ($12 \times 1 = 12$ Marks)

#	Question	Correct Answer	Explanation
1	The prefix nano represents a multiplier of:	B	10^{-9} . nano = one-billionth. Option A (10^{-6}) is micro, C (10^{-12}) is pico, and D (10^9) is giga.
2	Screw gauge, pitch 1 mm, 100 circular divisions. Least count:	B	0.01 mm . Least count = pitch $\div$ number of circular-scale divisions = 1 mm $\div$ 100 = 0.01 mm.
3	The area under a velocity-time graph represents the:	B	Distance covered . Area = velocity $\times$ time = distance. The <i>slope</i> (not the area) of a velocity-time graph gives acceleration, which is the trap in option A.
4	Car: rest to 30 m/s in 6 s. Acceleration:	A	5 m/s² . $a = (v - u)/t = (30 - 0)/6 = 5 \text{ m/s}^2$. Option B divides the wrong way; C and D multiply instead of dividing.
5	The SI unit of impulse is:	B	N·s (newton-second). Impulse = force $\times$ time = change in momentum, so its unit equals that of momentum (kg·m/s = N·s). N·m is the unit of torque/work; J is energy.
6	Body of mass 2 kg moving at 5 m/s. Momentum:	A	10 kg·m/s . $p = mv = 2 \times 5 = 10 \text{ kg·m/s}$. Option B divides; C squares the speed; D adds the numbers.
7	Centripetal force on a body moving in a circle is directed:	C	Towards the centre of the circle. “Centripetal” means “centre-seeking.” A force away from the centre (B) is the mistaken “centrifugal” idea; the tangent (A) is the direction of velocity, not the force.
8	Smallest friction for the same surfaces:	C	Rolling friction . Rolling friction is much smaller than sliding (and static) friction, which is why wheels and ball bearings are used to reduce friction.
9	Force 200 N on area 0.5 m ² . Pressure:	B	400 Pa . $P = F/A = 200/0.5 = 400 \text{ Pa}$. Option A divides the wrong way (0.5/200 path); D multiplies incorrectly.
10	Work done by a force is zero when the angle is:	C	90° . $W = Fd \cos\theta$; at $\theta = 90^\circ$, $\cos 90^\circ = 0$, so $W = 0$ (e.g. carrying a bag horizontally while gravity acts downward). At 0° work is maximum.
11	Body of mass 2 kg moving at 3 m/s. Kinetic energy:	B	9 J . $E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 2 \times 3^2 = \frac{1}{2} \times 2 \times 9 = 9 \text{ J}$. Option C forgets the $\frac{1}{2}$; A uses v instead of v^2 .
12	A concise, often mathematical, statement of a relationship always seen in nature is a:	C	Law (e.g. Newton’s law of motion). A <i>hypothesis</i> (A) is an untested assumption; a <i>theory</i> (B) explains <i>why</i> something happens, while a law states <i>what</i> happens.

SECTION B — Short Answer Key ($11 \times 3 = 33$ Marks)

2(i) — Pitch, least count, and screw-gauge calculation 3 marks

Pitch: the distance the screw moves forward (or backward) along the main scale in one complete rotation of the circular scale. 1

Least count: the smallest length that can be measured = pitch ÷ number of divisions on the circular scale. 1

$$\text{Least count} = \text{pitch} / \text{circular divisions} = 1 \text{ mm} / 50 = 0.02 \text{ mm} [1]$$

OR

(a) **0.0067 m** has **2** significant figures (leading zeros are not significant). 1

(b) **30500 J** has **3** significant figures (trailing zeros with no decimal point are not counted). 1

Rounding **4.5678** to three significant figures gives **4.57** (the digit after 6 is 7, so round up). 1

2(ii) — Scientific notation and its use 3 marks

$$0.000045 \text{ m} = 4.5 \times 10^{-5} \text{ m}. 1$$

$$6,400,000 \text{ m} = 6.4 \times 10^6 \text{ m}. 1$$

Use: scientific notation writes very large or very small numbers compactly, making them easier to read, compare, and use in calculations (and it shows significant figures clearly). 1

OR

$10^6 = \text{mega (M)}$; $10^{-3} = \text{milli (m)}$; $10^{-9} = \text{nano (n)}$. 2 (1 for any two correct, 2 for all three)

$$5 \text{ km} = 5 \times 10^3 \text{ m} = 5000 \text{ m}. 1$$

2(iii) — Distance, displacement, average speed 3 marks

(a) Total distance = 300 m + 100 m = 400 m [1] (b) Net displacement = 300 m (E) - 100 m (W) = 200 m towards East [1] (c) Total time = 20 s + 10 s = 30 s Average speed = total distance / total time = 400 / 30 = 13.3 m/s [1]

OR

Average speed = total distance ÷ total time; it is a single value describing the whole journey.

Instantaneous speed is the speed at one particular instant (the reading a speedometer shows at that moment). 1

$$180 \text{ km in } 2 \text{ h: Average speed} = 180/2 = 90 \text{ km/h} = 90 \times 1000/3600 = 25 \text{ m/s} [2]$$

2(iv) — Newton's second law and mass calculation 3 marks

Newton's second law: the acceleration of a body is directly proportional to the net force acting on it and inversely proportional to its mass, in the direction of the force; $F = ma$. 2

$$m = F / a = 24 \text{ N} / 3 \text{ m/s}^2 = 8 \text{ kg} [1]$$

OR

Inertia: the tendency of a body to resist any change in its state of rest or of uniform motion. 1

When a moving bus brakes suddenly, the lower body of the passenger stops with the seat, but the upper body tends to **keep moving forward** because of its **inertia of motion**. With no equal forward stopping force on the upper body, the passenger lurches forward. 2

2(v) — Conservation of momentum (skater and ball) 3 marks

Law: the total momentum of an isolated system remains constant; $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$. 1

Initial momentum = 0 (both at rest) $0 = (60 \times V) + (2 \times 9)$ $60 V = -18$ $V = -0.3 \text{ m/s}$ The skater recoils at 0.3 m/s, opposite to the ball. [2]

OR

Change in momentum $\Delta p = m(v - u)$. Take the initial direction as positive, so $u = +20 \text{ m/s}$ and $v = -20 \text{ m/s}$.

$$\Delta p = 0.15 \times (-20 - 20) = 0.15 \times (-40) = -6 \text{ kg.m/s (magnitude } 6 \text{ kg.m/s)} [2] \text{ Impulse} = \text{change in momentum} = 6 \text{ N.s} [1]$$

2(vi) — Torque definition and calculation 3 marks

Torque (moment of force): the turning effect of a force about a pivot, equal to the force multiplied by its perpendicular distance from the pivot. Formula: $\tau = F \times d$; SI unit N·m. 2

$$\tau = F \times d = 50 \text{ N} \times 0.4 \text{ m} = 20 \text{ N.m} [1]$$

OR

Principle of moments: when a body is in equilibrium, the sum of clockwise moments about any pivot equals the sum of anticlockwise moments about the same pivot. 1

$$\text{Anticlockwise} = \text{clockwise} \quad 4 \text{ N} \times 30 \text{ cm} = 6 \text{ N} \times d \quad 120 = 6d \rightarrow d = 20 \text{ cm} [2]$$

2(vii) — Centripetal force / orbital velocity 3 marks

Centripetal force: the net force directed towards the centre of a circle that keeps a body moving along a circular path; $F_c = mv^2/r$. 1

$$F_c = m v^2 / r = 0.5 \times (4)^2 / 1 = 0.5 \times 16 = 8 \text{ N} [2]$$

OR

Orbital velocity: the constant speed needed by a satellite to stay in its circular orbit around a planet. 1

$$v = 2 \pi r / T = 2 \times 3.14 \times (4.2 \times 10^7) / 86400 = 2.639 \times 10^8 / 86400 = 3054 \text{ m/s (about 3.05 km/s)} [2]$$

2(viii) — Hooke's law, spring constant, calculation 3 marks

Hooke's law: within the elastic limit, the extension produced in a spring is directly proportional to the applied force ($F = kx$). 1

Spring constant (k): the force needed to produce unit extension; SI unit N/m. 1

$$k = F / x = 12 \text{ N} / 0.04 \text{ m} = 300 \text{ N/m (4 cm = 0.04 m)} [1]$$

OR

The force-extension graph is a **straight line through the origin** up to the elastic limit, then it bends. 1

The **slope of the straight portion equals the spring constant k** (force per unit extension). 1 Beyond the elastic limit the spring is permanently deformed and does **not return** to its original length when the force is removed. 1

2(ix) — Pressure definition and floor pressure 3 marks

Pressure: the force acting normally (perpendicularly) per unit area of a surface; $P = F/A$; SI unit pascal (Pa), where $1 \text{ Pa} = 1 \text{ N/m}^2$. 2

$$P = F / A = 600 \text{ N} / 1.5 \text{ m}^2 = 400 \text{ Pa} [1]$$

OR

$$P = \rho g h = 1000 \times 10 \times 3 = 30000 \text{ Pa} = 30 \text{ kPa} [2]$$

Liquid pressure depends on (1) the **depth h** and (2) the **density of the liquid ρ** . It does not depend on the shape or area of the container. 1

2(x) — Work / power definitions and calculation 3 marks

Work: the product of force and the distance moved by the body in the direction of the force; $W = Fd$; SI unit joule (J). 2

$$F = W / d = 250 \text{ J} / 10 \text{ m} = 25 \text{ N} [1]$$

OR

Power: the rate of doing work (work done per unit time); $P = W/t$; SI unit watt (W), where $1 \text{ W} = 1 \text{ J/s}$. 2

$$P = W / t = 6000 \text{ J} / 30 \text{ s} = 200 \text{ W} [1]$$

2(xi) — Steps of the scientific method / role of physics 3 marks

Main steps, in order: (1) **observation** of a problem or phenomenon; (2) forming a **hypothesis** (a possible explanation); (3) performing **experiments** to test the hypothesis; (4) **analysing the results** and drawing a conclusion; (5) if confirmed, forming a **theory or law** (and reporting so others can reproduce it). 3 (about ½ mark per correct step, in correct order)

OR

Physics provides the basic principles on which technology is built; engineers turn these principles into useful devices. 1

Examples: (1) the study of **electromagnetism** led to electric generators and motors; (2) the study of **semiconductors / electronics** led to computers and mobile phones. (Also acceptable: optics → cameras and lenses; nuclear physics → medical scanners.) 2

SECTION C — Long Answer Key (4 × 5 = 20 Marks)

3(i) — Acceleration, distance, and v-t graph area 5 marks

Acceleration: the rate of change of velocity of a body; $a = (v - u)/t$; SI unit m/s^2 . 1

(a) $a = (v - u)/t = (30 - 10)/8 = 20/8 = 2.5 \text{ m/s}^2$ [1] (b) distance $s = ((u + v)/2) \times t = ((10 + 30)/2) \times 8 = 20 \times 8 = 160 \text{ m}$ [2] (check: $s = ut + \frac{1}{2} a t^2 = 10 \times 8 + 0.5 \times 2.5 \times 64 = 80 + 80 = 160 \text{ m}$)

(c) The **area under a velocity-time graph equals the distance covered** by the body. 1

OR

Three equations of motion (uniform acceleration): 2

(1) $v = u + at$ (2) $S = ut + \frac{1}{2}at^2$ (3) $2aS = v^2 - u^2$.

Symbols: u = initial velocity, v = final velocity, a = acceleration, t = time, S = distance covered. (Dropped from rest: $u = 0$, $a = g = 10 \text{ m/s}^2$, $t = 3 \text{ s}$.)

(a) $v = u + g t = 0 + 10 \times 3 = 30 \text{ m/s}$ [1] (b) $h = u t + \frac{1}{2} g t^2 = 0 + \frac{1}{2} \times 10 \times (3)^2 = 0.5 \times 10 \times 9 = 45 \text{ m}$ [2]

3(ii) — Momentum, conservation, and a collision 5 marks

Momentum: the quantity of motion of a body, equal to mass × velocity ($p = mv$); it is a vector. 1

Law of conservation of momentum: in an isolated system (no external force), the total momentum before an interaction equals the total momentum after it. 1

Before: $p = m_1 u_1 + m_2 u_2 = 1500 \times 12 + 1000 \times 0 = 18000 \text{ kg.m/s}$ [1] They lock together (combined mass = 2500 kg): $18000 = (1500 + 1000) \times v = 2500 v$ $v = 18000 / 2500 = 7.2 \text{ m/s}$ [2] (in the direction the first car was moving)

OR

Momentum = mass × velocity ($p = mv$). 1

Rocket propulsion: burning fuel produces hot gases that are ejected backward at high speed, giving the gases a large **backward momentum**. 1 Since the system's total momentum must stay constant (initially zero), the rocket gains an equal **forward momentum**, pushing it forward. 2 As fuel burns, the rocket's mass keeps decreasing while the gases keep pushing, so the same forward momentum produces a higher velocity and the rocket **speeds up**. 1

3(iii) — Pascal's principle, hydraulic lift / $P = \rho gh$ 5 marks

Pascal's principle: pressure applied to an enclosed fluid is transmitted equally and without loss in all directions throughout the fluid. **1**

Hydraulic lift: a small force on the small piston creates a pressure that is transmitted through the fluid to the large piston. Because the pressure is the same on both pistons, the larger area produces a much larger force (force multiplication). **2**

Pressure is equal: $F_1/A_1 = F_2/A_2$ $F_2 = (A_2 / A_1) \times F_1 = (0.5 / 0.01) \times 40 = 50 \times 40 = 2000 \text{ N}$ [2]

OR

Derivation of $P = \rho gh$: consider a liquid column of height h and base area A . Volume = $A \times h$; mass = density $\times$ volume = ρAh ; weight (force on base) = mass $\times g = \rho Ahg$. Pressure = force $\div$ area = $\rho Ahg / A = \rho gh$. **3**

Liquid pressure = $\rho gh = 1000 \times 10 \times 5 = 50000 \text{ Pa}$ [1] Total pressure = atmospheric + liquid = $1.0 \times 10^5 + 0.5 \times 10^5 = 1.5 \times 10^5 \text{ Pa}$ [1]

3(iv) — Power and efficiency / conservation of energy 5 marks

Power: the rate of doing work; $P = W/t$; SI unit watt (W). **Efficiency:** the ratio of useful output energy to total input energy, expressed as a percentage; $\eta = (\text{useful output} / \text{total input}) \times 100\%$. **1**

(a) Efficiency = (useful output / input) $\times 100 = (4000 / 5000) \times 100 = 80\%$ [1.5] (b) Useful power = useful work / time = $4000 \text{ J} / 25 \text{ s} = 160 \text{ W}$ [1.5]

(c) No real machine is 100% efficient because some input energy is **always lost as heat (and sound) due to friction and air resistance**, so the useful output is always less than the input. **1**

OR

Law of conservation of energy: energy can neither be created nor destroyed; it only changes from one form to another, so the total energy of a system stays constant. **1**

(a) PE at top = $mgh = 3 \times 10 \times 5 = 150 \text{ J}$ [1.5] (b) At the ground all PE has become KE: KE = 150 J [1] (c) $\frac{1}{2}mv^2 = 150$ $v^2 = (2 \times 150) / 3 = 100$ $v = 10 \text{ m/s}$ [1.5]